UNIVERSITÀ DI TRENTO
Department of Information Engineering and Computer Science

Automated Reasoning and **Formal Verification**

Laboratory 8

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Outline

1. Model Properties
   - Invariants
   - LTL
   - CTL
2. Fairness Constraints
3. Modelling a Program in nuXmv
4. Examples
5. Homework

Model Properties [1/2]

Specifying Properties

- directly **in the module**

```
LTLSPEC G  (req -> F  sum = op1 + op2);
```

- or **via nuXmv interactive shell**

```
nuXmv > check_ltlspec -p "G (req -> F sum = op1 + op2)"
```

- `show_property` lists all properties specified (in the module or via `add_property`):
- properties can be verified one at a time using its **database index**

```
nuXmv > show_property
**** PROPERTY LIST [ Type, Status, Counter-example Number, Name ] ****
--------------------------  PROPERTY LIST  -------------------------
000 :out < 2
  [Invar          False          1       N/A]
001 : G ( F out = 1)
  [LTL            Unchecked      N/A     N/A]
nuXmv > check_ltlspec -n 1
```

Model Properties [2/2]

Property verification:

- Each property is verified independently
- The result is either "`TRUE`" or "`FALSE` + counterexample"

Property types

Different kinds of properties are supported:

Invariants	: properties on every reachable state;
LTL	: properties on the computation paths;
CTL	: properties on the computation tree.

Invariants

- Invariant properties are specified via the keyword `INVARSPEC`:

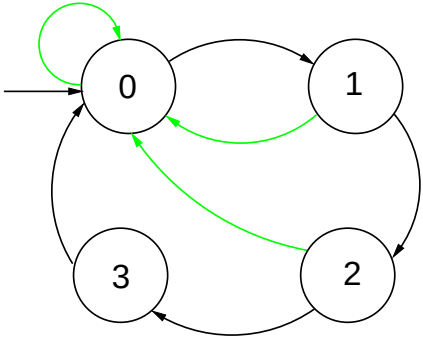
  `INVARSPEC <simple_expression>;`
- Invariants are checked via the `check_invar` command

Remark

When checking invariants, all the **fairness conditions** of the model are **ignored**.

Example: Modulo 4 Counter with Reset [1/2]

```
MODULE main
VAR  b0 : boolean; b1 : boolean;
     reset : boolean;
ASSIGN
  init(b0) := FALSE;
  next(b0) := case reset  : FALSE;
                  !reset : !b0;
              esac;
  init(b1) := FALSE;
  next(b1) := case reset : FALSE;
                  TRUE  : ((!b0 &  b1) |
                           ( b0 & !b1));
              esac;
DEFINE out := toint(b0) + 2 * toint(b1);
INVARSPEC out < 2;
```



Example: Modulo 4 Counter with Reset [2/2]

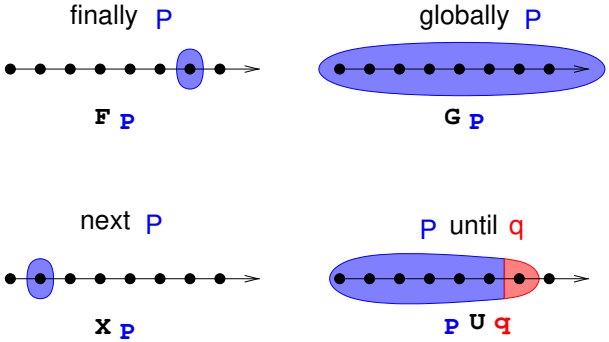
- The invariant is false

```
nuXmv > read_model -i counter4reset.smv
nuXmv > go; check_invar
-- invariant out < 2  is false
...
  -> State: 1.1 <-
    b0 = FALSE
    b1 = FALSE
    reset = FALSE
    out = 0
  -> State: 1.2 <-
    b0 = TRUE
    out = 1
  -> State: 1.3 <-
    b0 = FALSE
    b1 = TRUE
    out = 2
```

LTL Specifications

- LTL properties are specified via the keyword `LTLSPEC`:

  `LTLSPEC <ltl_expression>;`



- LTL properties are checked via the `check_ltlspec` command

LTL specifications

Specifications Examples:

- A state in which `out=3` is eventually reached

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  ```
  LTLSPEC F out = 3;
  ```

- Condition `out=0` holds until `reset` becomes false

LTL specifications

Specifications Examples:

- A state in which out=3 is eventually reached

```
LTLSPEC F out = 3;
```

- Condition out=0 holds until reset becomes false

```
LTLSPEC (out = 0) U (!reset);  -- False: reset can be true forever
LTLSPEC (!reset)  V (out = 0); -- True (V stands for "release")
```

- Every time a state with out=2 is reached, a state with out=3 is reached afterward

LTL specifications

Specifications Examples:

- A state in which out=3 is eventually reached

```
LTLSPEC F out = 3;
```

- Condition out=0 holds until reset becomes false

```
LTLSPEC (out = 0) U (!reset);  -- False: reset can be true forever
LTLSPEC (!reset)  V (out = 0); -- True (V stands for "release")
```

- Every time a state with out=2 is reached, a state with out=3 is reached afterward

```
LTLSPEC G (out = 2 -> F out = 3);
```

LTL specifications

All the previous specifications are false:

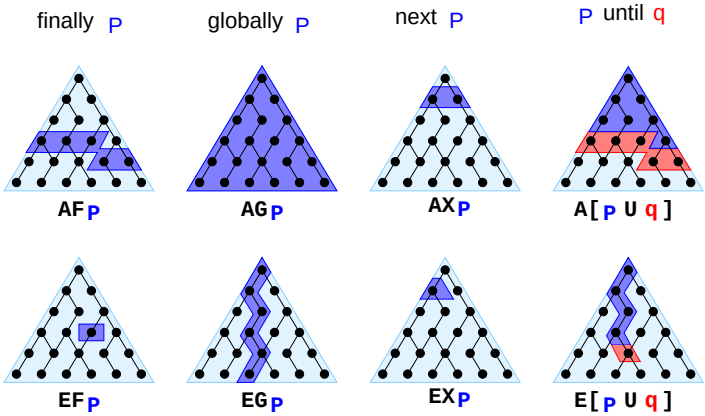
```
nuXmv > check_ltlspec
-- specification  F out = 3 is false ...
-- loop starts here --
-> State 1.1 <-
    b0 = FALSE
    b1 = FALSE
    reset = TRUE
    out = 0
-> State 1.2 <-
-- specification (out = 0 U (!reset)) is false ...
-- loop starts here --
-> State 2.1 <-
    b0 = FALSE
    b1 = FALSE
    reset = TRUE
    out = 0
-> State 2.2 <-
-- specification  G (out = 2 ->  F out = 3) is false ...
```

Q: Why?

CTL specifications

- CTL properties are specified via the keyword `CTLSPEC`:

```
CTLSPEC <ctl_expression>;
```



- CTL properties are checked via the `check_ctlspec` command

CTL specifications

Specifications Examples:

- It is possible to reach a state in which `out=3`

CTL specifications

Specifications Examples:

- It is possible to reach a state in which `out=3`

  ```
  CTLSPEC EF out = 3;
  ```

- It is inevitable that `out=3` is eventually reached

CTL specifications

Specifications Examples:

- It is possible to reach a state in which `out=3`

  ```
  CTLSPEC EF out = 3;
  ```

- It is inevitable that `out=3` is eventually reached

  ```
  CTLSPEC AF out = 3;
  ```

- It is always possible to reach a state in which `out=3`

CTL specifications

Specifications Examples:

- It is possible to reach a state in which `out=3`

  ```
  CTLSPEC EF out = 3;
  ```

- It is inevitable that `out=3` is eventually reached

  ```
  CTLSPEC AF out = 3;
  ```

- It is always possible to reach a state in which `out=3`

  ```
  CTLSPEC AG EF out = 3;
  ```

- Every time a state with `out=2` is reached, a state with `out=3` is reached afterward

CTL specifications

Specifications Examples:

- It is possible to reach a state in which out=3

  ```
  CTLSPEC EF out = 3;
  ```

- It is inevitable that out=3 is eventually reached

  ```
  CTLSPEC AF out = 3;
  ```

- It is always possible to reach a state in which out=3

  ```
  CTLSPEC AG EF out = 3;
  ```

- Every time a state with out=2 is reached, a state with out=3 is reached afterward

  ```
  CTLSPEC AG (out = 2 -> AF out = 3);
  ```

- The reset operation is correct

CTL specifications

Specifications Examples:

- It is possible to reach a state in which out=3

  ```
  CTLSPEC EF out = 3;
  ```

- It is inevitable that out=3 is eventually reached

  ```
  CTLSPEC AF out = 3;
  ```

- It is always possible to reach a state in which out=3

  ```
  CTLSPEC AG EF out = 3;
  ```

- Every time a state with out=2 is reached, a state with out=3 is reached afterward

  ```
  CTLSPEC AG (out = 2 -> AF out = 3);
  ```

- The reset operation is correct

  ```
  CTLSPEC AG (reset -> AX out = 0);
  ```

Outline

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2. Fairness Constraints
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The need for Fairness Constraints

The specification `F out = 3;` is not verified

- On the path where **reset** is always **1**, the system loops on a state where **out** $=$ **0**:
  `reset = TRUE, TRUE, TRUE, TRUE, TRUE, ...`
  `out = 0, 0, 0, 0, 0, ...`

Similar considerations for other properties:

- `F out = 1;`
- `F out = 2;`
- `G (out = 2 -> F out = 3);`
- `...`

The need for Fairness Constraints

The specification `F out = 3;` is not verified

- On the path where **reset** is always **1**, the system loops on a state where **out** $= \mathbf{0}$:
  `reset = TRUE, TRUE, TRUE, TRUE, TRUE, ...`
  `out = 0, 0, 0, 0, 0, ...`

Similar considerations for other properties:

- `F out = 1;`
- `F out = 2;`
- `G (out = 2 -> F out = 3);`
- `...`

Fairness

It would be **fair** to consider only paths in which the **counter** is **not reset** with such a high frequency, so as to hinder its desired functionality.

Fairness Constraints

nuXmv supports both **justice** and **compassion** fairness constraints:

Fairness: `JUSTICE <p>` consider only the executions that satisfy `p` **infinitely often**

Strong Fairness: `COMPASSION (<p>, <q>)` consider only those executions that either satisfy `p` **finitely often** or satisfy `q` **infinitely often**
(i.e., p is true infinitely often $\Longrightarrow$ q is true infinitely often)

Fairness Constraints

nuXmv supports both **justice** and **compassion** fairness constraints:

Fairness: `JUSTICE <p>` consider only the executions that satisfy p **infinitely often**

Strong Fairness: `COMPASSION (<p>, <q>)` consider only those executions that either satisfy p **finitely often** or satisfy q **infinitely often**
(i.e., p is true infinitely often $\Longrightarrow$ q is true infinitely often)

Remarks

- For **verification**, properties must hold only on **fair paths**
- When checking invariants, all the fairness conditions are **ignored**.
- Currently, compassion constraints have some limitations, since they are supported only for BDD-based LTL model checking.

Example: Modulo 4 Counter with Reset

Add the following fairness constraint to the model:

```
JUSTICE out = 3;
```

We consider only paths in which the counter reaches value 3 infinitely often

All the properties are now verified:

```
nuXmv > reset
nuXmv > read_model -i counter4reset.smv
nuXmv > go
nuXmv > check_ltlspec
-- specification F out = 1  is true
-- specification G (out = 2 -> F out = 3)  is true
-- specification G (reset -> F out = 0)  is true
```

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Example: model programs in nuXmv [1/4]

Q: Given the following **code**, how can we **model** and **verify** it with nuXmv?

```
def gcd(a: int, b: int) -> int:
  while a != b:
    if a > b:
      a = a-b
    else:
      b = b-a
  return a;  # at this point: GCD=a=b
```

Main idea

- We define a program counter pc that stores the current status of the execution (i.e., the line we reached).
- According to cycle and the conditional instructions, the program counter and the variables (when required) will change.

Example: model programs in nuXmv [2/4]

Step 1: label the **entry point** and the **exit point** of every block with the line number

```
def gcd(a: int, b: int) -> int:
  while a != b:   # l1
    if a > b:     # l2
      a = a-b     # l3
    else:
      b = b-a     # l4
  return a;       # l5 -- at this point: GCD=a=b
```

Example: model programs in nuXmv [3/4]

Step 2: encode the transition system with the assign style

```
MODULE main
VAR
  a: 0..100;  b: 0..100;
  pc: {l1, l2, l3, l4, l5};
ASSIGN
  init(pc) := l1;
  next(pc) := case
    pc = l1 & a != b  : l2;
    pc = l1 & a  = b  : l5;
    pc = l2 & a  > b  : l3;
    pc = l2 & a <= b  : l4;
    pc = l3 | pc = l4 : l1;
    pc = l5           : l5;
  esac;
```

```
next(a) := case
  pc = l3 & a > b : a - b;
  TRUE            : a;
esac;

next(b) := case
  pc = l4 & b >= a : b-a;
  TRUE             : b;
esac;
```

Example: model programs in nuXmv [4/4]

Step 2: (alternative): use the constraint style

```
MODULE main
VAR
  a : 0..100;  b : 0..100;  pc : {l1, l2, l3, l4, l5};
INIT pc = l1
TRANS pc = l1 -> (((a != b & next(pc) = l2) | (a = b & next(pc) = l5)) &
                  next(a) = a & next(b) = b);
TRANS pc = l2 -> (((a > b  & next(pc) = l3) | (a < b & next(pc) = l4)) &
                  next(a) = a & next(b) = b);
TRANS pc = l3 -> (next(pc) = l1 & next(a) = (a - b) & next(b) = b);
TRANS pc = l4 -> (next(pc) = l1 & next(b) = (b - a) & next(a) = a);
TRANS pc = l5 -> (next(pc) = l5 & next(a) = a       & next(b) = b);
```

Model programs in nuXmv: properties

Step 3: check the properties of the program

- Let's check if, given $a = 16$ and $b = 12$, then we will eventually get as a result 4.

Model programs in nuXmv: properties

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- Let's check if, given $a = 16$ and $b = 12$, then we will eventually get as a result 4.

```
LTLSPEC (a = 16 & b = 12) -> F (a = 4 & b = 4);
```

Model programs in nuXmv: properties

Step 3: check the properties of the program

- Let's check if, given $a = 16$ and $b = 12$, then we will eventually get as a result 4.
  `LTLSPEC (a = 16 & b = 12) -> F (a = 4 & b = 4);`
- Let's check if both numbers will never reach negative values:

Model programs in nuXmv: properties

Step 3: check the properties of the program

- Let's check if, given $a = 16$ and $b = 12$, then we will eventually get as a result 4.

```
LTLSPEC (a = 16 & b = 12) -> F (a = 4 & b = 4);
```

- Let's check if both numbers will never reach negative values:

```
INVARSPEC a > 0 & b > 0;
```

Outline

1. Model Properties

2. Fairness Constraints

3. Modelling a Program in nuXmv

4. Examples
   - Chemical reactions
   - The snail dungeon

5. Homework

Exercise 8.1

Assume the following chemical reactions hold:

$$
\begin{array}{rcrcr}
 & & 2O & \rightarrow & O_2 \\
C & + & O & \rightarrow & CO \\
2C & + & O_2 & \rightarrow & 2CO \\
C & + & O_2 & \rightarrow & CO_2
\end{array}
$$

Given 6 carbon atoms and 6 oxygen atoms, is there any way for the contents of this reaction vessel to progress to a state where it contains three molecules of CO2? Model the contents of the reaction vessel in nuXmv.

Science Modelling: variables [1/2]

- We can store the number of atoms and molecules in the current iteration with bounded integers.
- An enum variable can be used to define what reaction should be considered in the next step, ensuring non-determinism when necessary.

Science Modelling: variables [2/2]

```
MODULE main
VAR
  o        : 0..32;
  o2       : 0..32;
  c        : 0..32;
  co       : 0..32;
  co2      : 0..32;
  r : {r1, r2, r3, r4, none};
ASSIGN
  init(o)        := 6;
  init(c)        := 6;
  init(co)       := 0;
  init(co2)      := 0;
  init(o2)       := 0;
  init(r) := none;
```

Science Modelling: transitions [1/2]

Transitions to define the next reaction that will take place on the next step.

```
TRANS
  (next(o)  < 2)                -> (next(reaction) != r1);
TRANS
  (next(o)  < 1 | next(c) < 1) -> (next(reaction) != r2);
TRANS
  (next(o2) < 1 | next(c) < 2) -> (next(reaction) != r3);
TRANS
  (next(o2) < 1 | next(c) < 1) -> (next(reaction) != r4);
```

Science Modelling: transitions [2/2]

Transitions to define the new values for each molecule after a reaction took place.

```
TRANS
  r=none -> (next(o)  = o      & next(o2)  =  o2     & next(c) = c     &
             next(co) = co     & next(co2) = co2)
TRANS
  r=r1   -> (next(o)  =  o - 2 & next(o2)  =  o2 + 1 & next(c) = c     &
             next(co) = co     & next(co2) = co2);
TRANS
  r=r2   -> (next(o)  =  o - 1 & next(o2)  =  o2     & next(c) = c - 1 &
             next(co) = co + 1 & next(co2) = co2);
TRANS
  r=r3   -> (next(o)  = o      & next(o2)  =  o2 - 1 & next(c) = c - 1 &
             next(co) = co + 2 & next(co2) = co2);
TRANS
  r=r4   -> (next(o)  = o      & next(o2)  =  o2 - 1 & next(c) = c - 1 &
             next(co) = co     & next(co2) = co2 + 1);
```

Science Modelling: property

- If we are interested in knowing if there is a path that generates 3 CO_2 molecules, LTL apparently seems ineffective...
- ... but we can use it to search a valid counter-example corresponding to the desired execution.
- We try to verify that the number of CO_2 molecules does not reach 3 in any path:
  `LTLSPEC G (co2!=3)`
- If the property is not satisfied, we get a sequence of steps reaching a state where co2=3.

The Snail Dungeon

Exercise 8.2

You want to simulate the gameplay of "**The Snail Dungeon**":

- You have a path with 10 cells, with 2 good and 2 bad teleports. Use a variable `turn` whose value could be `{DICE, GOOD, BAD}`, and a variable `steps` counting how many times a dice has been thrown. Once set, the teleport positions remain fixed.
- Each turn you **throw a 3-valued dice** and **move forward** the designated number of cells. If the arrival cell is empty, you move on to the next turn. Notice that the dice **cannot get the same number from two consecutive throws**.
- If you get into a *good* teleport, the next value of turn will be `GOOD` and you will **move onward by 2 cells without increasing** `steps`.
- If you get into a *bad* teleport, the next value of turn will be `BAD` and you will **move back by 2 cells without increasing** `steps`.

Encode the game in nuXmv and find a way to reach position 10 with less than 3 steps.

The Snail Dungeon: variables

- The text already indicates some variables. Notice that with the default **BDD-based engine** we can only use **bounded integers**.

```
VAR
  pos   : 1..10;
  turn  : {DICE, GOOD, BAD};
  steps : 0..100;
  dice  : 1..3;
```

- We can use two arrays of two integers in `1..10` to store the positions of teleports.

```
good : array 0..1 of 1..10; bad  : array 0..1 of 1..10;
```

- These arrays are **fixed**. Instead of adding

```
ASSIGN next(good[0]) := good[0]; next(good[1]) := good[1]; ...
```

we can declare them as `FROZENVAR`:

```
FROZENVAR good : array 0..1 of 1..10;
          bad  : array 0..1 of 1..10;
```

The Snail Dungeon: initial values

- Initialize the game:

```
ASSIGN
  init(pos)   := 1;
  init(steps) := 0;
  init(turn)  :=DICE;
```

- And constrain the teleport positions:

```
INIT
  good[0] != good[1] & bad[0]  != bad[1] &
  good[0] != bad[0]  & good[0] != bad[1] &
  good[1] != bad[0]  & good[1] != bad[1];
```

The Snail Dungeon: transitions [1/2]

We encode the moves' logic:

```
ASSIGN
  next(turn) := case
    next(pos) = good[0] | next(pos) = good[1] : GOOD;
    next(pos) = bad[0]  | next(pos) = bad[1]  : BAD;
    TRUE: DICE;
  esac;
  next(steps) := case
    turn = DICE : min(steps + 1, 100);
    TRUE        : steps;
  esac;
  next(pos) := case
    turn = DICE : min(pos + dice, 10);
    turn = GOOD : min(pos + 2, 10);
    turn = BAD  : max(pos - 2, 1);
  esac;
```

Notice: we must ensure that bounded integers do not exceed their limits!

The Snail Dungeon: transitions [2/2]

Finally, we encode the dice logic: it cannot give the same value twice in a row.

```
TRANS
  (turn  = DICE -> next(dice) != dice) &
  (turn != DICE -> next(dice)  = dice);
```

The Snail Dungeon: properties

To get a trace where we reach position 10 with less than 3 steps, can negate the property and get a counter-example:

```
INVARSPEC pos=10 -> steps >= 3;
```

```
Trace Type: Counterexample
-> State: 1.1 <-
  good[0] = 8
  good[1] = 4
  bad[0] = 10
  bad[1] = 2
  pos = 1
  turn = DICE
  steps = 0
  dice = 3
-> State: 1.2 <-
  pos = 4
  turn = GOOD
  steps = 1
  dice = 2
-> State: 1.3 <-
  pos = 6
  turn = DICE
-> State: 1.4 <-
  pos = 8
  turn = GOOD
  steps = 2
  dice = 1
-> State: 1.5 <-
  pos = 10
  turn = BAD
```

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Homework

Homework 8.1: Bubblesort

Implement a transition system which sorts the following input array $\{4, 1, 3, 2, 5\}$ with increasing order. Verify the following properties:

- there exists no path in which the algorithm ends
- there exists no path in which the algorithm ends with a sorted array

Bubblesort

Bubblesort

You might use the following *bubblesort* code as reference:

```
def bubbleSort(A : list[int]) -> None:
  n = len(A)
  swapped = True
  while swapped:
    swapped = false
    for i in range(1, n):
      if A[i-1] > A[i]: # if this pair is out of order
        # swap and remember something changed
        A[i-1], A[i] = A[i], A[i-1]
        swapped = true
```